Skills Review

Exercise No. 13

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Course/Section: CPE4B **Date: Oct. 7, 2026**

Machine Problem:

Create a program using Interrupt Function 10h that will display standard TV color bars following the standard sizes. Use string character printing and cursor positioning to display your favorite International or Local TV Network information in the color bars. Graph the coordinates by drawing the expected output. Save the program using Exer13.asm as the filename.

Program:

```
; AARON LUDWIG A. ALTAR
; CPE4B

.model small
.stack
.data

var1    DB 'EST. 2026','$'
var2    DB 'A L T A R   D E V   S T U D I O','$'
var3    DB 'Code . Create . Innovate','$'

.code
start:

        ; screen background (0,0 - 24,79)
        mov ah, 06h             ; Function 06h - scroll up window
        mov bh, 1Eh             ; blue background, yellow text
        mov ch, 0               ; start row
        mov cl, 0               ; start column
        mov dh, 24              ; end row
        mov dl, 79              ; end column
        int 10h                 ; BIOS video interrupt - display the window

        ; badge shadow (3,26 - 18,57)
        mov ah, 06h
        mov bh, 00h             ; black
        mov ch, 3
        mov cl, 26
        mov dh, 18
        mov dl, 57
        int 10h
```

```
        ; badge outer ring (2,24 - 17,55)
        mov ah, 06h
        mov bh, 30h             ; cyan
        mov ch, 2
        mov cl, 24
        mov dh, 17
        mov dl, 55
        int 10h

        ; badge inner (3,26 - 16,53)
        mov ah, 06h
        mov bh, 70h             ; white, black text
        mov ch, 3
        mov cl, 26
        mov dh, 16
        mov dl, 53
        int 10h

        ; <---- corner accents ---->
        ; top-left accent (2,24 - 2,25)
        mov ah, 06h
        mov bh, 60h             ; yellow
        mov ch, 2
        mov cl, 24
        mov dh, 2
        mov dl, 25
        int 10h

        ; top-right accent (2,54 - 2,55)
        mov ah, 06h
        mov bh, 20h             ; green
        mov ch, 2
        mov cl, 54
        mov dh, 2
        mov dl, 55
        int 10h

        ; bottom-left accent (17,24 - 17,25)
        mov ah, 06h
        mov bh, 50h             ; magenta
        mov ch, 17
        mov cl, 24
        mov dh, 17
        mov dl, 25
        int 10h

        ; bottom-right accent (17,54 - 17,55)
```

```
        mov ah, 06h
        mov bh, 40h             ; red
        mov ch, 17
        mov cl, 54
        mov dh, 17
        mov dl, 55
        int 10h

        ; <---- letter A ---->
        ; A - top (5,29 - 6,39)
        mov ah, 06h
        mov bh, 40h             ; red
        mov ch, 5
        mov cl, 29
        mov dh, 6
        mov dl, 39
        int 10h

        ; A - left leg (7,29 - 14,31)
        mov ah, 06h
        mov bh, 40h             ; red
        mov ch, 7
        mov cl, 29
        mov dh, 14
        mov dl, 31
        int 10h

        ; A - right leg (7,37 - 14,39)
        mov ah, 06h
        mov bh, 40h             ; red
        mov ch, 7
        mov cl, 37
        mov dh, 14
        mov dl, 39
        int 10h

        ; A - crossbar (9,32 - 10,36)
        mov ah, 06h
        mov bh, 40h             ; red
        mov ch, 9
        mov cl, 32
        mov dh, 10
        mov dl, 36
        int 10h

        ; <---- letter L ---->
        ; L - vertical (5,42 - 14,44)
```

```
        mov ah, 06h
        mov bh, 10h             ; blue
        mov ch, 5
        mov cl, 42
        mov dh, 14
        mov dl, 44
        int 10h

        ; L - foot (13,45 - 14,50)
        mov ah, 06h
        mov bh, 10h             ; blue
        mov ch, 13
        mov cl, 45
        mov dh, 14
        mov dl, 50
        int 10h

        ; name banner (19,20 - 21,59)
        mov ah, 06h
        mov bh, 4Fh             ; red, white text
        mov ch, 19
        mov cl, 20
        mov dh, 21
        mov dl, 59
        int 10h

        mov ax, @data           ; initialize the data segment
        mov ds, ax

        mov ah, 02h             ; Function 02h - set cursor position
        mov bh, 00h             ; page 0
        mov dh, 16              ; row
        mov dl, 35              ; column
        int 10h
        mov ah, 09h             ; Function 09h - display string
        mov dx, offset var1     ; year under the letters
        int 21h

        mov ah, 02h
        mov bh, 00h
        mov dh, 20
        mov dl, 24
        int 10h
        mov ah, 09h
        mov dx, offset var2     ; logo name in the banner
        int 21h
```

```
        mov ah, 02h
        mov bh, 00h
        mov dh, 23
        mov dl, 28
        int 10h
        mov ah, 09h
        mov dx, offset var3     ; tagline
        int 21h

        mov ah, 02h             ; move the cursor near the bottom so the
        mov bh, 00h             ; DOS prompt will not cover the design
        mov dh, 23
        mov dl, 0
        int 10h

        mov ah, 4Ch             ; terminate program
        int 21h

end start
```

Output:

